SIGNAL SUPPORT IN COMBINE ARMS OPERATIONS
STUDENT HANDOUTS

TERMINAL LEARNING OBJECTIVE:

At the end of the subject, the students will be able to determine signal support in Combined Arms Operations IAW PAM 6-0102. (C3)

ENABLING LEARNING OBJECTIVES:

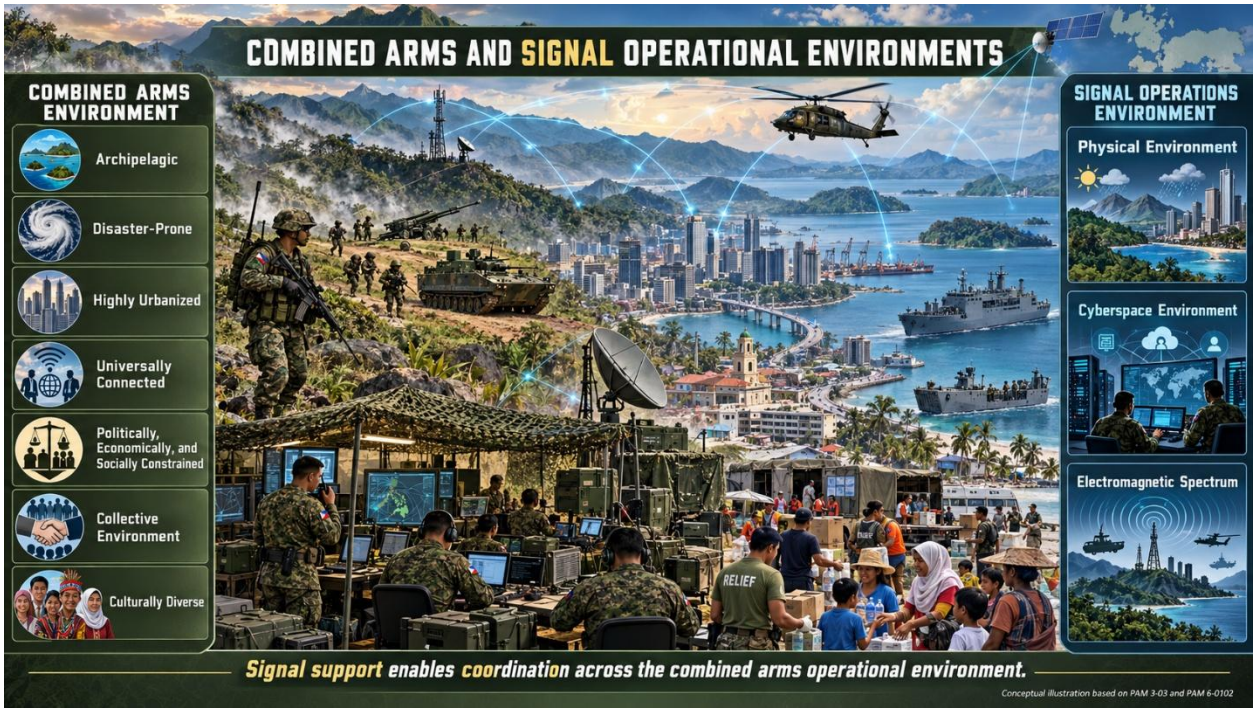
1. Discuss the roles of Signal Support in Military Operations. (C3)
2. Determine Signal Support Systems and Technologies. (C3)
3. Discuss Principles of Signal Support in Combined Arms Operations. (C3)
4. Discuss Tactical Considerations in Signal Support Operations. (C3)
5. Determine the Plan and Coordinate Signal Support in Combined Arms Operations. (C3)
6. Determine the security measures and protocols in Signal Support Operations. (C3)

Signal Operations	The synchronized actions of the Army Signal Corps to connect Army units and enable the transmission, storage, and security of information within Philippine Army infrastructure in support of Army operations.
Signal Support Operations	The provision of communication and information services that enable commanders to exercise command and control. These services include command and control communication, cyberspace operations and electronic warfare, signal repair and maintenance, and signal recovery and restoration.
Combined Arms Operations	An approach to warfare which seeks to integrate the different capabilities of an army to achieve mutually complementary effects capable of overwhelming enemy forces. It is the synchronization, integration and seamless execution of Army capabilities to achieve desired effects.
Signal Support to Combined Arms Operations	The application of those signal services to connect and enable coordination among infantry, armor, artillery, engineers, and other participating capabilities, supporting their synchronized and integrated employment toward mission accomplishment.

Signal Operations and Signal Support Operations are closely connected activities. Combined Arms Operations as the integration and synchronization of complementary capabilities, while signal services that enable communication and command and control.

Signal Operations connects the Army; Signal Support Operations provides and sustains the services behind that connection; Signal Support to Combined Arms Operations applies those services so the different combat arms can work together.

Combined Arms Operations and Signal Operations describe the operational environment from different perspectives: Combined Arms considers the conditions affecting the employment of forces, while Signal considers the environments in which communication, electronics, and cyber systems operate.



1. Operational Environment in Combined Arms Operations

The operational environment are the conditions, circumstances, and factors that affect the employment of capabilities and the commander's decisions. The seven characteristics of the Philippine Army's operational environment:

Characteristic	Meaning and effect on Combined Arms Operations
Archipelagic	Islands, mountains, coastlines, and bodies of water affect maneuver, transportation, logistics, and coordination of forces.
Disaster-Prone	Natural calamities require combined arms units to undertake disaster response and humanitarian assistance alongside their warfighting responsibilities.
Highly Urbanized	Buildings, obstacles, civilians, and protected facilities affect movement and the employment of firepower.
Universally Connected	Interconnected communication and information networks enable mission command and a common operational picture, while creating vulnerabilities.
Politically, Economically, and Socially Constrained	National priorities, social conditions, and resource limitations influence capability development and operations.

Collective Environment	Operations involve cooperation with government agencies, nongovernment organizations, and partner nations.
Culturally Diverse	Different traditions, beliefs, and practices must be considered to establish legitimacy and gain public support.

Commanders analyze these conditions using PMESII-PT—Political, Military, Economic, Social, Information, Infrastructure, Physical Environment, and Time—and METT-TC—Mission, Enemy, Terrain and Weather, Troops and Support Available, Time Available, and Civil Considerations.

2. Operational Environment in Signal Operations

Environment	Meaning	Effect on Signal Support
Physical Environment	The natural and man-made surroundings where signal systems operate, including islands, mountains, forests, buildings, and weather.	Affects communication coverage, signal propagation, equipment placement, and the survival of communication assets.
Cyberspace Environment	The Army's hardware, software, networks, and people who interact with those systems. It includes physical, logical, and people domains.	Requires functioning and protected networks, correct configurations, and responsible users to support information exchange.
Electromagnetic Spectrum Environment	The range of electromagnetic frequencies used by systems such as tactical radios, microwave links, cellular systems, and wireless networks.	Enables wireless communication and requires consideration of interference and electronic warfare threats.

Signal units provide connectivity across the physical, cyberspace, and electromagnetic environments so that combined arms forces can coordinate their capabilities under the seven conditions described above.

For Example; in an urban operation, buildings affect troop movement and radio coverage; networks support the exchange of information; and the electromagnetic spectrum enables radio communication between maneuver units, fire support, and headquarters.

ELO 1: Discuss the roles of Signal Support in Military Operations

Signal Support enables commanders and units to communicate, coordinate their actions, and exchange information so they can accomplish the mission. It also protects, maintains, and restores the systems used for that communication.

Signal Operations organizes Signal Support into **four main service categories**.

1. **Enable command and control**

Command and control mean the commander directs the forces under their authority to accomplish a mission. Signal Support provides the communication systems that make this possible.

Through these systems, commanders issue orders, subordinate units submit reports, and units coordinate with one another.

Three communication services:

Services	Simple meaning	Example
Fixed communication	Connects headquarters and established facilities.	Army telephone systems, PANET, email, and video teleconferencing.
Tactical communication	Connects operating troops while moving or halted.	HF, VHF, and UHF tactical radios and combat net radio networks.
Special purpose communication	Supplements fixed and tactical systems, particularly for routine or administrative communication.	Cellular phones, satellite phones, and commercial videoconferencing, subject to communication security policies.

Example: An infantry company reports its situation to the battalion headquarters. The commander receives the report and issues updated instructions through the communication system.

What you should remember: Signal Support provides the connection; the commander uses that connection to exercise command and control.

2. Support cyberspace operations and electronic warfare

Signal Support also operates in computer networks and the electromagnetic spectrum. In cyberspace, its roles include operating Army information networks and defending them against threats. The Information Network Operations and Defensive Cyber Operations as Signal Corps responsibilities. It describes Active Defense Operations as a collaborative effort with intelligence and civil-military operations units. For electronic warfare, the three broad objectives: support, attack, and protect.

Example: A headquarters relies on its computer network to exchange operational reports. Signal personnel keep the network functioning and help protect it against unauthorized access or disruption. In the electromagnetic environment, Signal Support contributes to the effective and protected use of communication systems.

What you should remember: Establishing communication is only part of the task. The information and systems must also be protected.

3. Keep equipment ready through repair and maintenance

Communication equipment must remain serviceable throughout military operations. The maintenance facilities and Signal Mobile Maintenance and Repair Teams (SMMART) that inspect equipment, conduct repairs, and provide refresher training to supported units.

Example: A unit's tactical radio develops a fault. Signal maintenance personnel inspect and repair it so the unit can continue communicating.

What you should remember: Maintenance sustains communication capability and helps prevent equipment problems from interrupting the mission.

4. Recover and restore disrupted communication

Signal Support must restore communication when equipment, facilities, or networks are damaged by disasters, attacks, or other incidents. The deployable teams for restoring tactical operations centers, command and control communications, and networks, as well as Cyber Security Incident Response Teams (CSIRT).

Example: A typhoon damages a headquarters' communication facilities. Signal teams deploy to repair or restore the connections so commanders can resume directing their units.

What you should remember: When communication fails, Signal Support helps bring it back.

How these roles support different military operations

Setting	**Role of Signal Support**
Combined Arms Operations	Enables the communication needed to coordinate infantry, armor, artillery, and other supporting capabilities. This is an application of the command and control role.
Joint Operations	Enables communication interoperability between the Army and other major services through compatible equipment and coordinated radio operators.
Coalition Operations	Establishes communication links with partner nations while maintaining each nation's sovereign network.
Humanitarian Assistance and Disaster Response	Connects military units and other response organizations and restores disrupted communications. The disaster response radio net to remain separate from the Army's secured operational net.

The framework to remember: Connect, Sustain, and Recover

- Connect: Establish connectivity among Army units.

- Sustain: Maintain connectivity throughout operations.
- Recover: Repair and restore communication when it is attacked or destroyed.

Signal Support connects the force, keeps it connected, and restores its connections so commanders and units can continue accomplishing the mission.

ELO 2: Determine Signal Support Systems and Technologies.

To determine Signal Support Systems and Technologies means to identify the communication and information systems a unit needs to accomplish its mission. Understand what each system does and when it is appropriate to use it. These systems under command-and-control communication, cyberspace operations and electronic warfare, maintenance, and restoration.

1. Fixed communication systems

Fixed communication systems connect headquarters and established military facilities. They form the backbone of the Army's voice and data communications.

System or technology	What it does
Philippine Army Bonifacio Telephone Exchange (BONTEX)	Connects Army telephone systems with higher headquarters.
Philippine Army Network (PANET)	Provides a secured, managed Army intranet for administrative and mission-critical information systems.
Voice over Internet Protocol (VoIP)	Carries telephone conversations through an IP network.
Email, video teleconferencing, and collaboration systems	Enable headquarters to exchange information and conduct coordination meetings.
Deployable Brigade Tactical Operations Center System (DBTOCS)	Extends fixed communication services to brigade headquarters.

Example: A brigade headquarters needs to send reports, access Army information systems, and attend a coordination meeting with higher headquarters. Fixed communication services support these requirements.

2. Tactical communication systems

Tactical communication systems connect troops conducting field operations. They support communication both on the move and on the halt. The voice communication remains the foundation of tactical communication.

Technology	What you should understand

High Frequency (HF) radios	One of the tactical radio technologies
Very High Frequency (VHF) radios	Another tactical radio technology used to connect operating units.
Ultra-High Frequency (UHF) radios	A further tactical radio technology supporting operational communication.
Combat Net Radio (CNR)	Provides radio networks through which operating units communicate.
Radio over IP (ROIP)	Links radio communications through IP networks.
Ad hoc network technologies	Support tactical networking among participating equipment.

Example: An infantry company moving away from its headquarters still needs to receive instructions and submit situation reports. Tactical radios provide the connection required during movement and after the unit halts.

Do not select a radio based only on its frequency category. Determine whether the available equipment can meet the mission's coverage, terrain, interoperability, and security requirements.

3. Special purpose communication systems

Special purpose communication supplements the Army's fixed and tactical systems.

This includes:

- Cellular phones
- Satellite phones
- Commercial videoconferencing platforms

These technologies support routine or administrative communication. Their use must follow existing communication security (COMSEC) policies.

Example: Personnel may use an authorized commercial platform for an administrative meeting. The convenience of that platform does not automatically make it suitable for sensitive operational information.

4. Cyberspace systems and technologies

Cyberspace systems allow the Army to process, store, and exchange digital information.

Domain	**Components**	**Simple explanation**
Physical	Computers, network devices, towers, and transmission facilities	The equipment that carries and processes information.
Logical	Network topology, firewall configurations, and internet protocols	The configurations and relationships that make the equipment work together.

People	Individuals interacting with Army networks through computers, tablets, and phones	The users and personnel who operate or interact with the systems.

Example: A headquarters computer cannot exchange reports merely because it is switched on. It needs a working network connection, correct configurations, and authorized users.

5. Technologies operating in the electromagnetic spectrum

The electromagnetic spectrum is the environment through which wireless systems transmit information. The tactical radio, microwave, cellular, and wireless computer networks as systems using this environment.

Signal planners must consider the spectrum because friendly and hostile forces both use it. The electronic warfare through three broad objectives: support, attack, and protect.

Example: A unit may have serviceable radios, but interference or hostile activity can still disrupt communication. Therefore, equipment availability alone does not guarantee a working connection.

6. Maintenance and restoration capabilities

A complete signal system also needs the capability to remain serviceable and recover from disruption.

Capability	**Purpose**
Signal Maintenance and Training Facilities (SMTF)	Conduct radio repairs and support equipment readiness.
Signal Mobile Maintenance and Repair Teams (SMMART)	Visit units to inspect equipment, conduct repairs, and provide refresher training.
DND-AFP electro-optic repair and maintenance facility	Supports the repair of night fighting systems.
Deployable Signal Teams and restoration teams	Restore command centers, communications, and networks after damage or disruption.
Cyber Security Incident Response Teams (CSIRT)	Provide a response capability for cybersecurity incidents.

These are support capabilities and organizations, rather than communication technologies themselves. They keep the technologies available for operations.

How do you determine what a unit needs?

Apply the Connect, Sustain, and Recover framework by asking:

1. Connect: Who must communicate, what information must they exchange, and which systems can connect them?

2. Sustain: What personnel, power, maintenance, protection, and support will keep those systems operating?
3. Recover: What alternate connections and restoration capabilities are available if communication fails?

For example, a field unit may need tactical radios to communicate during movement, a connection to Army information networks at its command post, and maintenance and restoration support to keep those services available. The exact combination depends on the mission and operational environment.

Determine the communication requirement first, then select and support the systems that can meet it throughout the operation.

ELO 3 Discuss the Principles of Signal Support in Combined Arms Operations

Signal Support in Combined Arms Operations must allow the different combat arms to communicate and coordinate throughout the mission. Infantry, armor, artillery, and other supporting elements need to exchange orders, reports, and requests so their capabilities work together.

The five fundamental principles of Signal Operations: Operations Focused, Interoperable, Redundant, Scalable, and Secured. These principles guide Signal Support across its services.

1. Operations Focused — Support the mission

Signal Support must be designed around what the operating force needs to accomplish. Communication support should never become the limiting factor in the commander's ability to execute a mission. Signal personnel must therefore consider the duration of operations, equipment readiness, power requirements, and connectivity.

Combined arms example: Infantry and armored units are moving together. Their communication support must allow them to remain connected during movement. A system that works only at the headquarters does not fully meet their operational requirement.

Remember: Start with the mission's communication needs, then determine the equipment and support required.

2. Interoperable — Enable units to communicate with one another

Interoperability means communication and information systems are compatible and follow standardized arrangements. Units within the Army—and units from other major services—to be able to communicate during operations.

Combined arms example: Infantry identifies an obstacle and needs engineer support. The infantry, engineers, and headquarters must have a workable means of exchanging the request and coordinating the response.

Owning radios alone does not guarantee interoperability. The systems and their operating arrangements must enable the required exchange of information. Remember: Check whether the units can actually communicate with each other.

3. Redundant — Provide backup connections

Redundancy means having additional communication paths and backup capabilities when a system fails. This includes multiple transmission paths, backups, self-healing strategies, and data replication.

Combined arms example: The main connection between a maneuver unit and headquarters fails. An established alternate communication path allows the unit to continue sending reports and receiving instructions.

Redundancy must address the connection, not merely the number of devices. Several devices depending on the same failed infrastructure may all lose service. Remember: Ask, “If this connection fails, how will the unit continue communicating?”

4. Scalable — Adjust support as the operation changes

Scalability means Signal Support can expand and be tailored to the size, situation, and phases of an operation. The networks that support command and control at home station, while en-route, and in deployed environments.

Combined arms example: A battalion receives additional armor, artillery, and engineer elements. Signal Support must accommodate the added users and coordination requirements.

Scalability also means tailoring support appropriately when the force reorganizes or moves into another phase. Remember: Communication support must adapt as the force and mission change.

5. Secured — Protect communication and information

Signal Support must comply with communication security, or COMSEC, policies regardless of the technology used. The advances in technology do not remove the requirement for security.

Combined arms example: Units exchange orders, locations, and operational reports. Their communication arrangements must protect that information according to the applicable COMSEC requirements.

A convenient communication platform must be assessed against those requirements before it is used for operational information. Remember: A working connection must also meet the required security standard.

Applying all five principles together
Imagine a combined arms force conducting an operation with infantry, armor, artillery, and engineers. Its Signal Support should meet these checks:

Principle	**Question to ask**
Operations Focused	Does communication support the mission, movement, and duration of operations?
Interoperable	Can all required elements exchange information with one another and headquarters?
Redundant	Can communication continue if a connection or system fails?
Scalable	Can support adapt when units are added or the operation changes?
Secured	Does the arrangement comply with COMSEC requirements?

These are instructional applications of the five principles of Signal Support. Together, they support the Connect, Sustain, and Recover framework: establish connectivity, maintain it throughout operations, and restore it after disruption.

ELO 4: Discuss Tactical Considerations in Signal Support Operations

Tactical considerations in Signal Support Operations are the field conditions and mission requirements that affect how we establish, protect, maintain, and restore communication. “Will our communication systems support the troops throughout the actual operation?” Having radios and computers is only the beginning. Addresses these considerations through the Signal Estimate and Signal Planning Process

1. Understand the situation using METAL
The METAL is a guide for preparing the Signal Estimate:

Factor	What you must determine	Simple example
M — Mission	What the unit must accomplish and what communication it requires.	A moving unit needs communication while traveling and after reaching its destination.
E — Enemy	Enemy signal capabilities and threats to friendly communication.	Consider possible interception, jamming, or cyberattacks.
T — Troops	Available personnel and their readiness to provide Signal Support.	Determine whether trained operators and technicians are available.
A — Area of Operations	Conditions in the area that affect communication.	Mountains, forests, buildings, and distances may affect coverage.
L — Logistical Requirements	Resources needed to support communication throughout the mission.	Consider power, replacement equipment, repair support, and transportation.

Remember: The Signal Estimate must be updated when the situation changes.

2. Consider terrain, weather, and signal coverage

The environment affects whether communication can reach the intended unit. The island separation, mountains, vegetation, urban structures, and storms create challenges for Signal Operations. Signal planners must therefore consider terrain and propagation conditions when locating communication facilities.

Example: A radio connection works at the command post but becomes unreliable when a unit moves behind a mountain. The planner must identify that potential coverage gap and arrange suitable support.

The site survey or reconnaissance and determination of signal coverage and gaps.

Lesson: Verify communication coverage against where troops will actually operate.

3. Consider troop movement and facility locations

Signal Support must follow the tactical disposition and projected operations of the supported force.

The communication facilities to be located with consideration of:

- Where supported units and support units are positioned.
- Terrain and signal propagation.
- Projected operations.
- Accessibility.
- Administrative and logistical requirements.

Example: An infantry element advances while supporting elements remain farther behind. Communication arrangements must continue connecting them as their positions change.

Lesson: A location that provides good coverage at the beginning may become unsuitable later.

4. Check equipment, personnel, and resources

Plan with the resources that are actually available and ready.

Resource	**What to check**
Equipment	What is on hand, serviceable, unserviceable, committed, or held in reserve—and what each system can do.
Personnel	Actual staffing, training, shortages, deployments, and reserves.
Funds	Support for installation, operation, maintenance, and administrative requirements.

Example: A unit has several radios, but some require repair and others lack trained operators. The usable communication capability is less than the equipment inventory suggests.

Lesson: Count usable capability, not just equipment.

5. Ensure interoperability and coordination

The units that must work together must also be able to communicate together. Signal planners obtain requirements from higher headquarters, subordinate units, adjacent units, and staff members. They also follow SOPs and the applicable Communication-Electronics Operating Instructions (CEOI) and Communication-Electronics Standing Instructions (CESI).

Combined arms example: Infantry, armor, artillery, and engineers need communication arrangements that allow their required coordination and reporting.

Lesson: Check connections across the force, including supporting and adjacent units.

6. Establish priorities when resources are limited

Communication requirements may exceed available capability.
The planners to compare requirements with resources and establish priorities based on supportability and importance to mission accomplishment. The commander's guidance determines the required means and priorities.

Example: If resources cannot support every requirement simultaneously, the planner identifies the shortfall and recommends support priorities consistent with the commander's intent.

Lesson: Allocate limited resources according to mission importance.

7. Prepare for communication failure and hostile disruption

A tactical communication plan must include what happens when the main connection fails.
The planning requires for:

- Interrupted connectivity: Redundant communication, repair, replacement, and technical support.
- Electronic warfare: Protection against jamming and interception.
- Cyberattacks: Network monitoring and incident response arrangements.
- Compromised frequencies: Procedures for changing frequencies when required.

It also calls for Primary, Alternate, Contingency, and Emergency (PACE) plans and alternate routing.

Example: A unit loses its primary connection. Personnel must already know which authorized backup arrangement to use and how to obtain restoration support.

Lesson: Prepare recovery arrangements before the disruption occurs.

8. Protect the systems and the information

Communication must remain secure throughout the operation. The compliance with COMSEC policies and inclusion of security measures in planning and execution.

Example: A commercially available device may provide connectivity, but its use must still meet the security requirements for the information being exchanged.

Lesson: Assess both whether the connection works and whether it is appropriate for the information.

Remember these three questions: Can we connect the force? Can we sustain that connection as the situation changes? Can we recover it after disruption? They apply the Connect, Sustain, and Recover framework to tactical conditions.

ELO 5: Determine Plan and Coordinate Signal Support in Combined Arms Operations.

To plan and coordinate Signal Support in Combined Arms Operations means to determine communication requirements, organize available resources, and arrange how all participating units will communicate throughout the mission.

Infantry, armor, artillery, engineers, and other supporting elements must exchange information so their capabilities work together. Your task as a signal planner is to establish the connections that support this coordination.

The Signal Estimate and Signal Planning Process, which work alongside the Military Decision-Making Process (MDMP) and, for small units, Troop Leading Procedures (TLP).

1. Understand the mission and prepare the Signal Estimate
First, determine the situation before selecting communication systems.
Use METAL to guide the Signal Estimate:

Factor	Question to answer
Mission	What must the force accomplish, and what communication will it need?
Enemy	What enemy signal capabilities or threats could affect our systems?
Troops	What personnel are available to provide Signal Support?
Area of Operations	How will terrain, distances, and other environmental conditions affect communication?
Logistical Requirements	What resources are needed to keep the systems operating?

Example: A combined arms force will move from an assembly area to another location. The estimate must consider communication during preparation, movement, and operations at the destination. The estimate is updated as the situation changes.

2. Determine who needs to communicate and why

Translate the commander's intent into specific communication requirements. The planner to obtain guidance from:

- The commander.
- Other staff members.
- Higher headquarters, subordinate units, and adjacent units.
- SOPs.
- CEOI and CESI, which provide operating instructions and explain their use.

For each requirement, identify who must exchange information, what information they need, and when they need the connection.

Combined arms example:

Element	Illustrative communication requirement
Infantry	Submit situation reports and receive instructions.
Armor	Coordinate movement with infantry and headquarters.
Artillery	Exchange support requests and coordination information through established procedures.
Engineers	Coordinate mobility support with maneuver units.
Logistics elements	Receive supply, maintenance, and transportation requirements.
Headquarters	Direct the force and maintain coordination with higher and adjacent units.

Lesson: A signal plan must connect the people and units whose coordination is necessary for the mission.

3. Determine available resources

Compare the requirements with what the unit can actually provide. The assessment of:

- Equipment: Availability, serviceability, maintenance readiness, reserves, and system capabilities.
- Personnel: Staffing, training, shortages, commitments, and reserves.
- Funds: Support for installation, operation, maintenance, and related requirements.

Example: The force needs several communication connections, but some equipment is unserviceable. The planner identifies the shortfall and determines whether repair, reallocation, or additional support is needed.

4. Establish support priorities

When resources are insufficient, prioritize requirements according to their importance to mission accomplishment. The planner compares requirements with capabilities and recommends priorities consistent with the commander's guidance.

Example: A shortage prevents simultaneous support to every requested service. The planner explains the limitations and recommends which requirements should receive support first.

Lesson: Priorities must reflect the mission and commander's intent.

5. Plan continuity and recovery

Determine how communication will continue or be restored when disrupted. The provisions for interrupted connectivity, electronic warfare, cyberattacks, and compromised communication frequencies. It also requires Primary, Alternate, Contingency, and Emergency (PACE) plans.

Arrangement	**Simple meaning**
Primary	The normal communication arrangement.
Alternate	A planned substitute if the primary arrangement fails.
Contingency	An additional option for a specified disruption or situation.
Emergency	The arrangement available when other planned options cannot meet the urgent requirement.

Example: A maneuver unit loses its primary connection. It follows the established backup arrangement while technical personnel work to restore service.
Lesson: Units must know the backup procedures before communication fails.

6. Prepare and coordinate the tentative plan

Bring the requirements, resources, priorities, and recovery arrangements into one practical plan. The planning for resource allocation, facility locations, security, interconnections, and alternate routing. Facility locations must consider terrain, propagation, unit positions, projected operations, accessibility, and logistics. Coordination means confirming that the supported units understand and can use these arrangements.

Coordinate with	**What to confirm**
Commander and operations staff	The plan supports the operational concept, priorities, and phases.
Higher headquarters	Required connections and available support.
Subordinate and supporting units	Their requirements, capabilities, operating arrangements, and responsibilities.
Adjacent units	Necessary coordination links.
Logistics and maintenance personnel	Resources for sustained operation, repair, and replacement.

Example: If the operations staff changes a unit's movement or location, the signal planner checks whether the communication coverage and support arrangements must also change.

8. Publish the plan and keep it updated

Communicate the approved arrangements through formal plans and orders. The preparation of the Signal Plan or C4S Annex to the Operations Order or Plan, including required diagrams, instructions, appendices, and tabs. As conditions change, update the Signal Estimate and coordinate necessary adjustments.

Planning determines how the force will communicate; coordination ensures all participating units understand and can carry out those arrangements. Throughout the process, apply Connect, Sustain, and Recover—establish the connections, keep them operating, and restore them after disruption.

ELO 6: Determine security measures and protocols in Signal Support Operations

Determining security measures and protocols in Signal Support Operations means identifying what must be protected, what threats could affect it, and which prescribed procedures personnel must follow. All communication services remain governed by communication security—or COMSEC—policy, regardless of the technology used. Security must be included in the signal plan and applied throughout the operational environment.

1. Understand what you are protecting

Signal Support protects the transmission, storage, and security of information exchanged between commanders and units.

Three things cyber threats can affect:

Security concern	**Simple meaning**	**Example**
Confidentiality	Information remains accessible only to authorized persons.	An unauthorized person must not obtain an operational report.
Integrity	Information remains accurate and is protected from unauthorized alteration.	A transmitted instruction must not be changed by an intruder.
Availability	Authorized users can access information and systems when needed.	Headquarters must be able to communicate with subordinate units during operations.

Remember: Security includes protecting information from disclosure, alteration, and disruption.

2. Follow COMSEC requirements

COMSEC provides the governing security requirements for communication. The commercial devices—such as cellular phones, satellite phones, and commercial videoconferencing platforms—must be used prudently and according to existing COMSEC policies.

Example: A messaging application may be convenient, but you must first determine whether its use is authorized for the information you intend to send. Learn and follow the applicable unit policies. It establishes the requirement to follow COMSEC; it does not provide every detailed security procedure.

3. Follow SOPs, CEOI, and CESI

Protocols are prescribed instructions that make communication consistent and controlled.

Sources of guidance:

Document	**Function**
Standard Operating Procedures (SOPs)	Provide established unit procedures for signal systems and activities.
Communication-Electronics Operating Instructions (CEOI)	Provide specific operating instructions—described as “what to do.”
Communication-Electronics Standing Instructions (CESI)	Explain the use of the CEOI—described as “how to do.”

Example: When a communication arrangement changes, operators follow the prescribed instructions so participating units make coordinated changes. Remember: Personnel must use the applicable instructions rather than invent their own procedures during an operation.

4. Protect networks against cyber threats

Defensive Cyber Operations protect PANET against cyber threats and provide a response to incidents.

The signal plans include:

- Network monitoring, including mobile ad hoc networks.
- Procedures for employing first responders.
- Procedures for employing the Cyber Security Incident Response Team (CSIRT).

The CSIRT maintains an incident response plan, investigates incidents, manages incident updates, remediates problems, and recommends improvements.

Example: Personnel notice suspicious activity affecting a headquarters network. They report it through the prescribed channels so authorized responders can investigate and act.

Remember: Cybersecurity requires both protective measures and an organized response when something goes wrong.

5. Protect communication against electronic threats

Tactical communication is vulnerable to jamming and interception. The plans to include measures that protect operating units from electronic attacks. It identifies frequency hopping and anti-jamming capabilities among electronic protection measures.

Example: If interference disrupts a radio connection, personnel use the authorized protection and backup procedures established for their equipment and unit.

Remember: These capabilities must be planned and coordinated; their use depends on available equipment and prescribed procedures.

6. Prepare for compromised equipment or frequencies

The communication equipment may fall into adversary hands, compromising operational radio frequencies. The signal plan must therefore include procedures for changing frequencies efficiently and quickly.

Example: If equipment is lost or captured, report the incident and follow the unit's prescribed compromise procedures. Authorized personnel determine and coordinate the required changes.

Remember: A compromise can affect the communication arrangements of several units, so the response must be coordinated.

7. Maintain communication during disruption

Availability is supported by redundancy and recovery arrangements. The calls for redundant communication, equipment repair and replacement, technical support, alternate routing, and Primary, Alternate, Contingency, and Emergency plans.

Example: If the primary connection fails, the unit uses its established backup arrangement while restoration personnel address the problem.

Remember: Backup arrangements must also comply with applicable security requirements.

8. Separate disaster-response communication from secured Army networks

For humanitarian assistance and disaster response, the response radio net to be different and separate from the Army's secured operational net. It considers the disaster-response net unclassified and unsecured.

Example: Army personnel may coordinate relief activities with civilian responders on the disaster-response net. Sensitive operational information must be handled through the appropriate authorized arrangements.